Classification

Pengenalan pola membutuhkan kekeraturan data.

Classification → Task to assign class labels by selecting from predefined class

Task type → Binary classification / Multiclass classification } Based on the size of predefined class

Task type → Single-label classification / Multilabel classification } Based on number of classes

Task type → Hard classification → output tides / Soft classification → output labels } Based on output : ~~class~~ class or probabilities

Formal definition

$f : \mathbb{R}^d \rightarrow L$ ⇒ Single class classification

$|L| = 2$: Binary classification

$|L| > 2$: Multiclass classification

$f : \mathbb{R}^d \rightarrow \mathbb{R}^L$ ⇒ Soft / fuzzy classification.

Apabila jika probabilitas dijumlah = 1

$f : \mathbb{R}^d \rightarrow 2^L$ ⇒ Multi-Label classification

$f : \mathbb{R}^d \rightarrow \{0,1\}^L$

$f : \mathbb{R}^d \rightarrow \mathbb{R}^L$ ⇒ Soft/fuzzy classification w/ Multi-Label [illegible]

bersifat independen, [illegible] dijumlah lebih [illegible]

Regression

↳ Linear regression is based on the linear relationship

↳ Dependent variable [illegible] → Response

$$\boxed{Y = \beta_0 + \beta_1 x}$$

```
>> from sklearn import linear_model
>> regr = linear_model.LinearRegression()
>> regr.fit(train_x, train_y)
```

Estimating Parameters : Least Square Estimator

↳ Objective function : Minimize sum of the squares of the residuals / errors (SSE)

⇒ SSE : Sum Squared Errors ≈> An accuracy measure where the errors are squared, then added

$$\boxed{SSE = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2}$$

$$\boxed{MAE = \frac{1}{n} \sum |y - \hat{y}|} \qquad \boxed{MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \tilde{y}_i)^2}$$

Target function is estimated by hypothesis as a linear function of x :

$h_\theta(x) = \theta_0 + \theta_1 x + \dots + \theta_n x_n = \sum \theta_i x_i$, i in $[0 \dots n]$ → $h_\theta(x) = \theta^T \cdot x$

Cost function $J(\theta)$ → >>> , bad predictions
→ <<< , good predictions

Supervised Learning

⇒ Learning from training set that is a set of input-output pairs $\{\langle x_i, y_i \rangle\}_{i \in \{1, \dots, m\}}$

↳ Part of inductive learning => Generate general function.

Y → Classification problem , Discrete set
→ Regression problem , Continuous set

AFM
31/08/2022
①

Linear discriminant ⇝ Kombinasi dari seluruh linear komponen x
↳ merupakan bentuk dari linear discriminant function.

Discriminant function:
$g(x) = w^T x$

$\boxed{w_i x_i}$ → digunakan u/ menentukan keputusan.
Hyperplane decision surface

Generalized Linear Discriminant function
↳ $\boxed{g(x) = \sum_{i=1}^{z} a_i y_i(x)}$ $\boxed{g(x) = a^T y}$

a : weight factor
$y_i(x)$ = arbitary function of (x)

SVM awalnya dirancang u/ binary classification.
↳ dikumukan secara matematis u/ mencari hyperplane yang optimal.
↳ mencari support vector => "Data terluar".
↳ Kernel memetakan data ke dimensi tinggi
↳ supaya tetap menjaga asumsi ~~linearly separable~~.
linearly separable.

Soft margin => margin bisa digeser, cocok pada kasus imbalance learning.

$\alpha > 0$ => Support vector

Breadth => "Habisin dulu yang selevel" } Queue
↳ First-in-first out

DFS => Last-in-first-out → Stack
↳ DFS-limited => Dibatasi kedalamannya. kalau kepanjangan balik lagi

$\boxed{\text{DFS Vs DFS-limited}}$

BFS

Informed search / Blind search
↳ Informed : BFS, DFS
↳ Informed search
↳ Heuristic search
↳ Local search

Informed search berisi "heuristik", atau perkiraan.
↳ use an evaluation function

Greedy best-first search.

A* search → Ideas: avoid expanding paths that are already expensive.
Evaluation search $f(n) = g(n) + h(n)$

PEAS: Performance measure, Environment, Actuator, Sensor

Environment type: Static vs Dynamic, Discrete vs Continuous, Deterministic vs Non-Deterministic.

Static: Environment berubah tidak selama AI mikir.

Deterministic: Aksi harus ditentukan dari state yang sekarang.

Leveling of agent: Problem solved agent, Knowledge based agent, learning agent.

Problem solved agent: semua state sudah diketahui oleh agent (Harus fully observable)

State space => Ruang kemungkinan state terjadi

Syarat PSA => Fully observable, Discrete, seluruh state space sudah diketahui.

IF3170

Mencari solusi terbaik.

Knowledge Based Agent:
- Tidak ada info state space
- Hanya tau kondisi atau "clue"

Learning Based Agent:
- Tidak ada info state space
- Tidak ada knowledge
- Knowledge dibuat berdasarkan observed data.

Problem Solving Agent

Agent design: Formulate problem → Search solution → execute.

↳ Satisfy goal

↳ 4 component: Initial state, operator, Goal test, path cost

Path cost → yang sudah dilalui ; Dipilih
→ diprediksikan.

Search:
Uninformed / Blind Search
↳ Example: DFS, BFS
Informed Search
↳ Heuristic Search
Local Search